Bayesian Fusion Forests

Heterogeneous treatment effects on survival from randomised and real-world data

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Two sources, two weaknesses

Randomised trial

✓ no confounding

× underpowered for sub-groups

× short follow-up

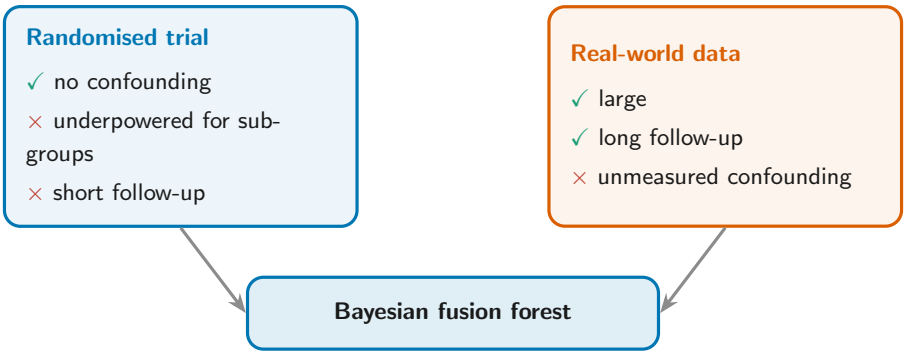
Real-world data

✓ large

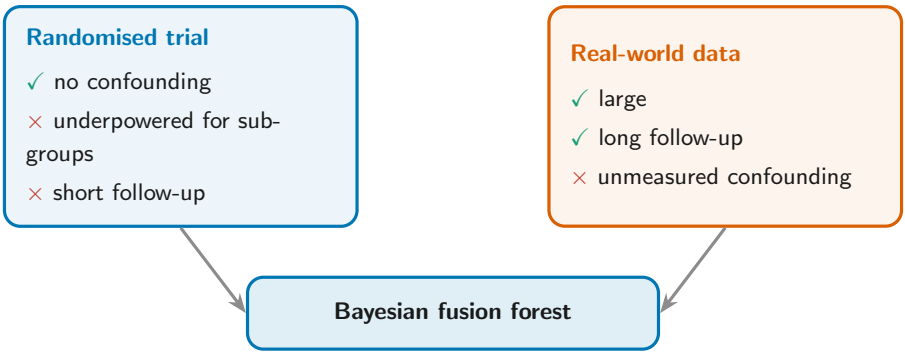
✓ long follow-up

× unmeasured confounding

Two sources, two weaknesses



Two sources, two weaknesses



Borrow the power, not the bias.

Motivating example: antiretroviral therapy for HIV

ACTG 175 — trial

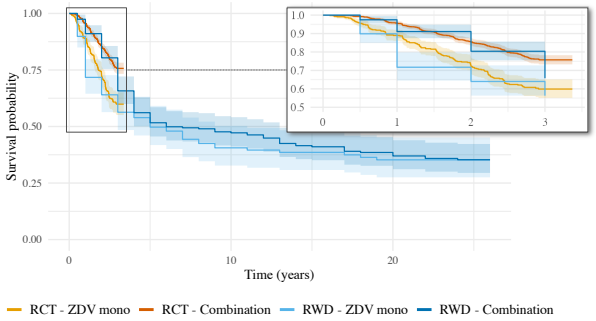
$n = 1771$, randomised, $\sim$3 years

MACS — cohort

$n = 373$, observational, $\sim$25 years, annual visits

Combination therapy vs. ZDV monotherapy.

Who benefits, and by how much?



The trial never reaches median survival.

Three obstacles

Unmeasured confounding

in the real-world data

Between-source heterogeneity

baselines differ

Censoring

right *and* interval

Three obstacles

Unmeasured confounding

in the real-world data

Between-source heterogeneity

baselines differ

Censoring

right *and* interval

Existing methods handle one or two.

We handle all three in one nonparametric model.

Contributions

- **Swap the assumption.** Drop unconfoundedness in the RWD; assume instead that the *treatment effect transports*.
- **Fusion for survival.** An AFT decomposition that takes right- *and* interval-censored outcomes.
- **The prior does the borrowing.** One tree ensemble per causal component, each regularised for its own role.
- **Errors may differ too.** A hierarchical Dirichlet process mixture: source-specific shape, shared components.

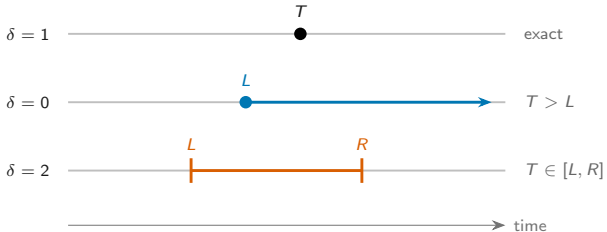
What we observe

Source $S \in \{0, 1\}$, treatment $A \in \{0, 1\}$, covariates $X \in \mathbb{R}^p$.

Survival time T observed only through a pair (L, R) with $T \in [L, R]$.

Trial: right-censored.

Cohort: interval-censored.



Estimand: the acceleration factor

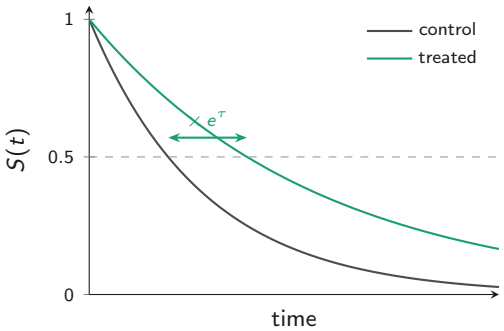
CATE on the log-time scale:

$$\tau(x) \;=\; \mathbb{E}\big[\log T(1) - \log T(0) \mid X = x\big].$$

On the time scale,

$$T(1) \mid X = x \;\overset{d}{=}\; e^{\tau(x)}\, T(0) \mid X = x.$$

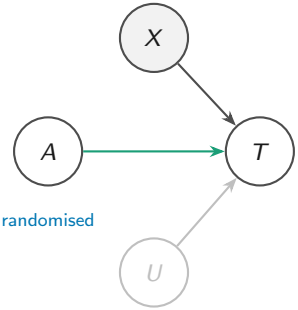
- Treatment *stretches the clock*.
- Collapsible; keeps its causal meaning where the hazard ratio does not.



$e^{\tau(x)} = 2 \Rightarrow$ median survival doubles.

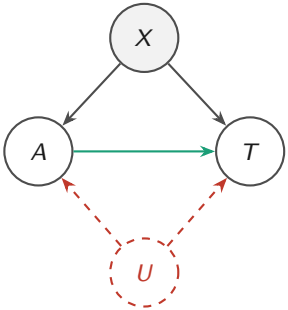
The assumption we drop, and the one we keep

Trial



No arrow into A: unconfounded by design.

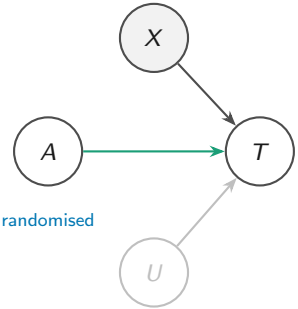
Real-world data



U unmeasured: back-door path stays open.

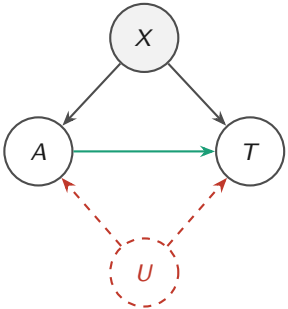
The assumption we drop, and the one we keep

Trial



No arrow into A: unconfounded by design.

Real-world data



U unmeasured: back-door path stays open.

Transportability. $\tau(x)$ is the same in both sources.

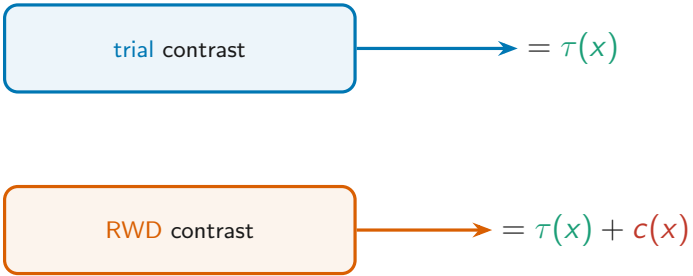
One decomposition, four functions

$$\log T = \underbrace{m_0^{\mathrm{sh}}(X)}_{\text{shared baseline}} + (1-S)\underbrace{d(X)}_{\text{source deviation}} + A\underbrace{\tau(X)}_{\text{treatment effect}} + (1-S)A\underbrace{c(X)}_{\text{confounding}} + \varepsilon$$

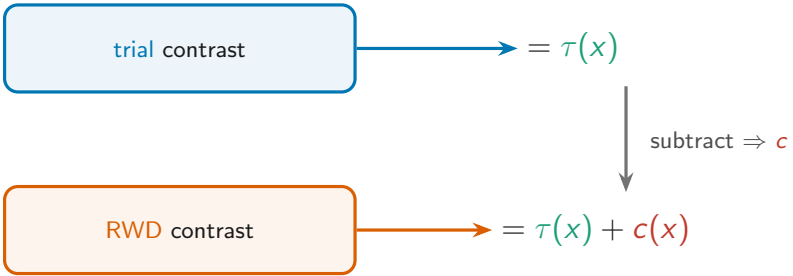
	control	treated
trial	m_0^{sh}	$m_0^{\mathrm{sh}} + \tau$
RWD	$m_0^{\mathrm{sh}} + d$	$m_0^{\mathrm{sh}} + d + \tau + c$

The confounding function c is the gap between the real-world contrast and the causal effect.

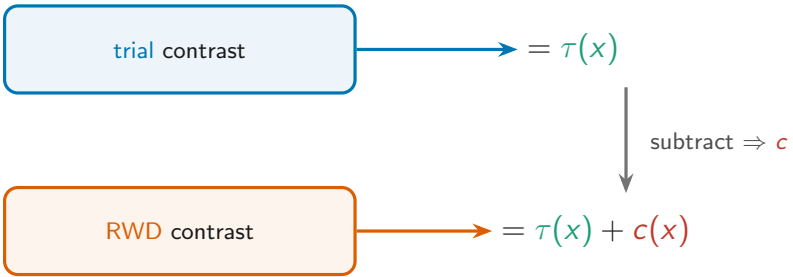
Identification: who pins down what



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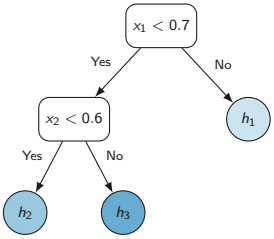
Identification: who pins down what



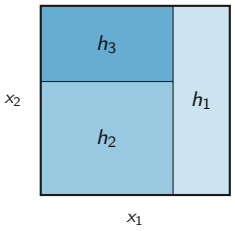
The RWD alone identifies only the sum.

Not a sample-size problem — no amount of real-world data resolves it.

BART in one slide



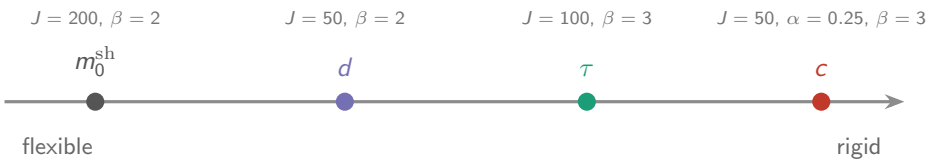




$$f(x) = \sum_{j=1}^{J} g(x; \mathcal{T}_j, \mathcal{H}_j)$$

A sum of J shallow trees. Depth-penalised prior $\alpha(1+d)^{-\beta}$; Gaussian step heights of scale k.

One forest per component — and a ladder of regularisation



- m_0^{sh} enters every observation ⇒ let it be complex.
- d is mean-zero ⇒ the prior *centres on equality* of the two baselines.
- τ shrunk towards a homogeneous effect.
- c shrunk towards a constant bias — it comes from what we did not measure.

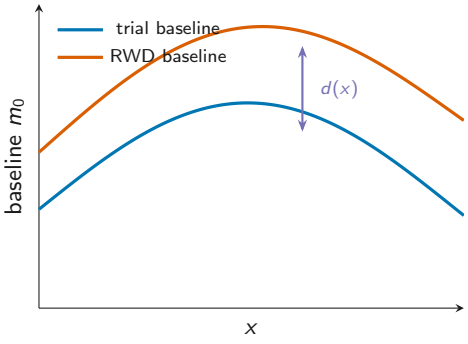
Borrowing is a prior, not a switch

$$m_0(X, S) \;=\; m_0^{\mathrm{sh}}(X) \;+\; (1 - S)\, d(X)$$

The deviation scale k_d tunes how far the two baselines may drift apart.

- $k_d \to 0$: full pooling.
- k_d large: two separate baselines.

A meta-analytic-predictive prior on the prognostic surface.



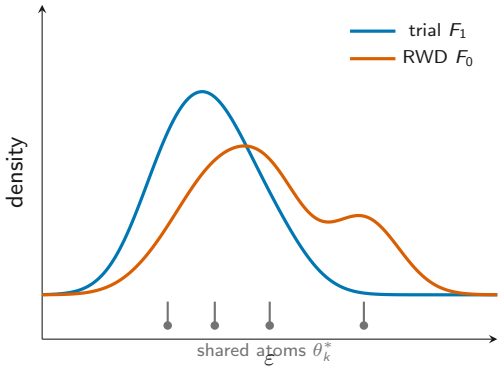
The errors may differ too

Protocol, follow-up quality and extreme outcomes differ between sources.

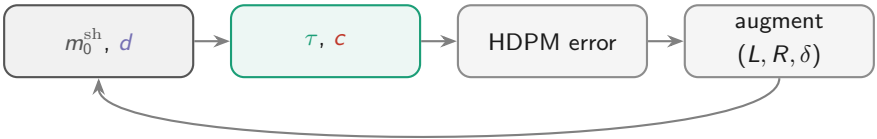
Hierarchical Dirichlet process mixture:

$$G^* \sim \mathrm{DP}(\gamma, H), \quad G_s^* \sim \mathrm{DP}(M_s, G^*).$$

- Shared mixture components.
- Source-specific weights.
- Centred, so $\mathbb{E}[\varepsilon \mid S] = 0$.



Posterior inference



- Blocked Gibbs: each forest updates against its partial residual, so every step is a *standard* BART update.
- Censored times are latent variables (Tanner & Wong, 1987) — this is where interval censoring enters, not the identification argument.
- Average effects by a hierarchical Bayesian bootstrap over the target covariate distribution.

Simulation design

$n_{\mathrm{RCT}} = 150$, $n_{\mathrm{RWD}} = 350$, $p = 10$, 1000 replications.

Trial right-censored ($\approx 35\%$);
RWD interval-censored at 8 visits.

Trial error Gaussian, RWD error skewed.

Two dials:

- λ_u — strength of the unmeasured confounding.
- λ_d — size of the between-source gap.

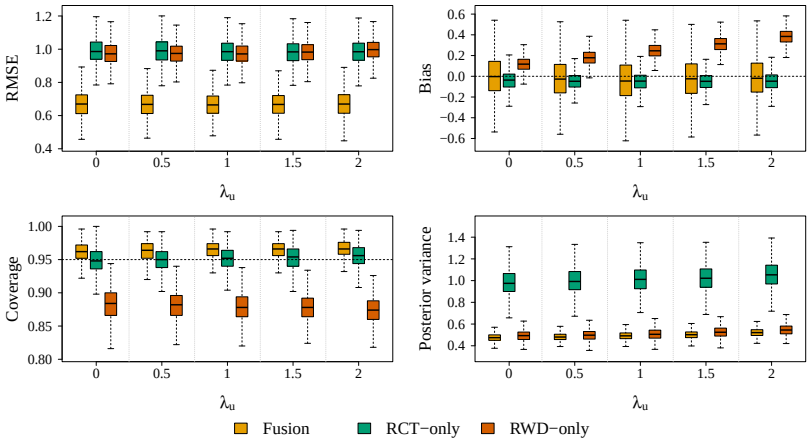
trial-only forest

RWD-only forest

fusion forest

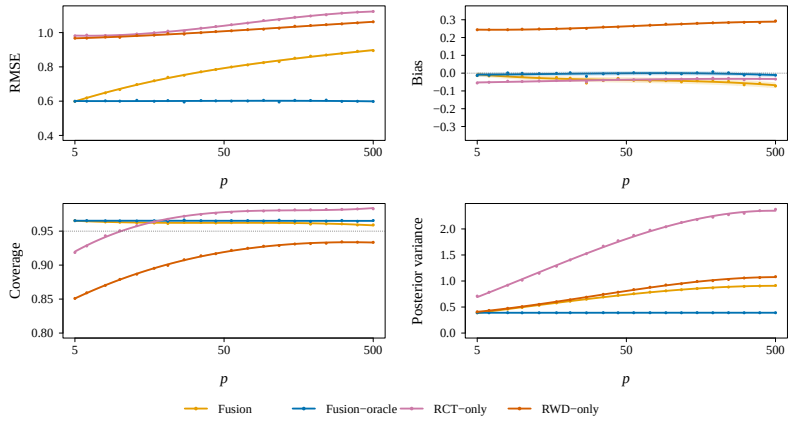
4 ML benchmarks
$\times$ S/T $\times$ trial/pool

Unbiased as the confounding grows



RWD-only bias grows with λ_u; fusion stays flat at half the posterior variance of the trial-only fit.

The gain survives a growing covariate space



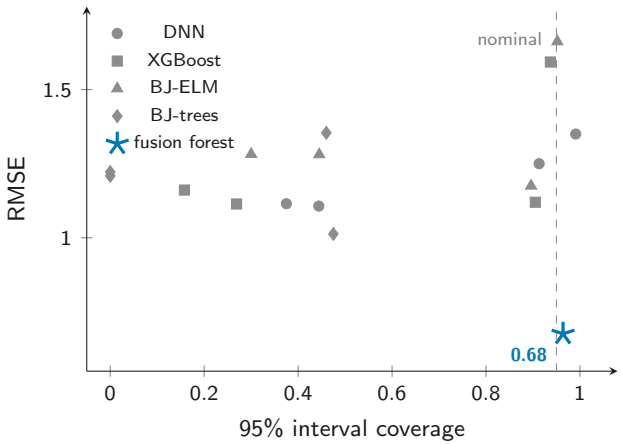
$p : 5 \to 500$, five active. RMSE ratio to trial-only rises from 0.68 to 0.81 — below one throughout.

Against flexible machine learning

Four predictors $\times$ {S, T} $\times$ {trial, pool}.

None targets a causal contrast; none can separate τ from c.

Only the fusion forest lands in the good corner: lowest RMSE *and* nominal coverage.

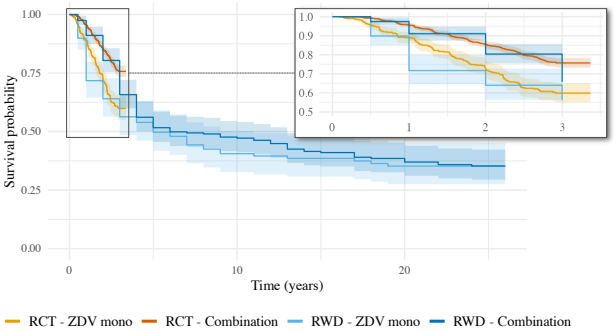


Combination therapy vs. ZDV monotherapy

Aligned population: men, baseline CD4 200–500.
Six baseline covariates.

ACTG 175: $n = 1771$, right-censored.

MACS: $n = 373$, interval-censored at annual visits.

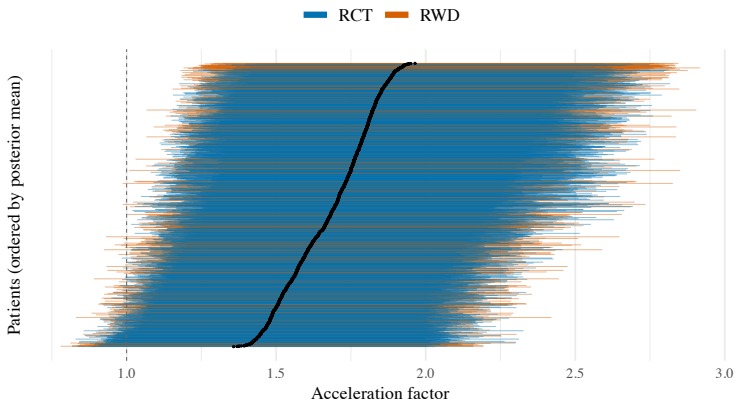


Average acceleration factor

$$\widehat{e^{\tau}} = 1.65 \quad [1.43;\ 1.90].$$

Trial-only: 1.70 [1.51; 1.92].

Every patient, with an interval



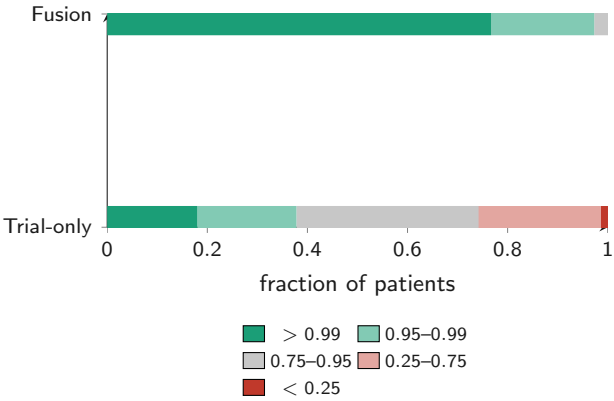
Subject-level acceleration factor, ordered by posterior mean. Average interval width falls from 3.16 to 1.21: a 62% reduction over the trial alone.

From inconclusive to near-certain

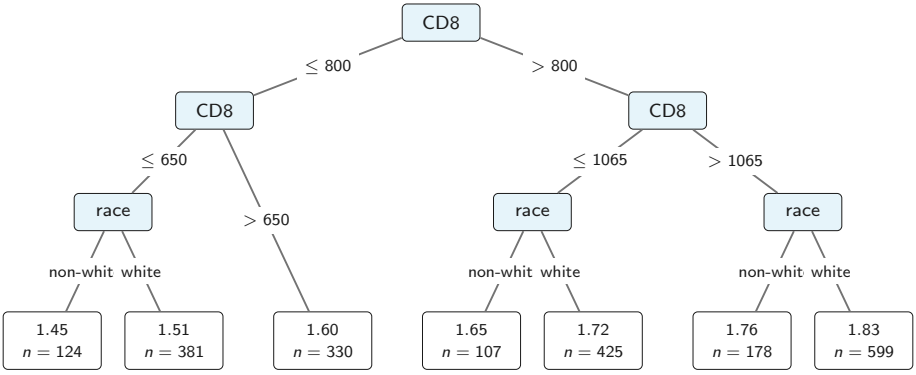
Posterior probability of benefit, $\mathbb{P}(\tau(x_i) > 0 \mid \text{data})$, over the *trial* population.

The trial alone leaves a quarter of patients inconclusive.

The fusion finds benefit for nearly everyone — with a slightly *smaller* average effect.



Who benefits most?



Posterior projection of τ onto a depth-three tree. Only CD8 and race enter; every leaf exceeds 1.4.

Take-aways

- **Trade one assumption for another.** Give up unconfoundedness in the real-world data; keep transportability of the effect. The confounding function absorbs the rest.

- **Put the borrowing in the prior.** A forest per causal component, each regularised for its role — shared baseline, mean-zero deviation, shrunken confounding.

- **It pays off.** Unbiased under confounding, half the posterior variance of a trial-only analysis, and benefit established for nearly every HIV patient where the trial alone was inconclusive.

Thanks for your attention!



Paper

Preprint on arXiv.



R package `FusionForests`

on GitHub; CRAN planned.

Stay up to date

`tijn-jacobs.github.io`

References

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Assumptions in full

1. **Consistency.** $A = a \Rightarrow T = T(a), \mathcal{C} = \mathcal{C}(a)$.
2. **Unconfoundedness of the RCT.** $T(a) \perp\!\!\!\perp A \mid X, S = 1$.
3. **Positivity.** $0 < \mathbb{P}(A = 1 \mid X = x, S = s) < 1$.
4. **Cross-source transportability.**
   $\mathbb{E}[\log T(1) - \log T(0) \mid X, S = 1] = \mathbb{E}[\log T(1) - \log T(0) \mid X, S = 0]$.
5. **Conditionally non-informative censoring.** $T(a) \perp\!\!\!\perp \mathcal{C}(a) \mid X, A, S$.

Assumptions 1, 3 and 5 are standard. Assumption 5 enters only the likelihood and the data augmentation, never the identification argument — which is why interval censoring costs nothing causally.

The confounding function

$$c(x) \;=\; \underbrace{\mathbb{E}[\log T \mid x, A=1, S=0] - \mathbb{E}[\log T \mid x, A=0, S=0]}_{\text{observed RWD contrast}} \;-\; \tau(x)$$

- Introduced as a sensitivity-analysis device within a single study (Robins, 2000).
- Here it is a *fitted* object: the trial identifies τ, so the RWD identifies c.
- Shrunk towards a constant: $(\alpha_c, \beta_c) = (0.25, 3)$, $J_c = 50$.

Full model

$$\log T_i = m_0^{\mathrm{sh}}(X_i) + (1 - S_i)\, d(X_i) + A_i\, \tau(X_i) + (1 - S_i) A_i\, c(X_i) + \varepsilon_i,$$

$$m_0^{\mathrm{sh}} \sim \mathrm{BART}(200,\ k_{\mathrm{sh}},\ 0.95,\ 2),$$

$$d \sim \mathrm{BART}(50,\ k_d,\ 0.95,\ 2),$$

$$\tau \sim \mathrm{BART}(100,\ k_\tau,\ 0.95,\ 3),$$

$$c \sim \mathrm{BART}(50,\ k_c,\ 0.25,\ 3),$$

$$\varepsilon_i \mid S_i = s \sim \int \sigma_s^{-1} \phi\big((w - \theta)/\sigma_s\big)\, dG_s(\theta),$$

$$G_s = \textstyle\sum_k \pi_{sk}\, \delta_{\theta_k^* - \mu_s},$$

$$G_s^* \mid G^*, M_s \sim \mathrm{DP}(M_s, G^*), \qquad G^* \mid \gamma \sim \mathrm{DP}(\gamma, H), \quad H = \mathcal{N}(0, \sigma_\theta^2),$$

$$\gamma \sim \mathrm{Gamma}(a_\gamma, b_\gamma), \qquad M_s \sim \mathrm{Gamma}(a_M, b_M), \qquad \sigma_s^2 \sim \nu\lambda/\chi_\nu^2.$$

Full comparison with machine-learning alternatives

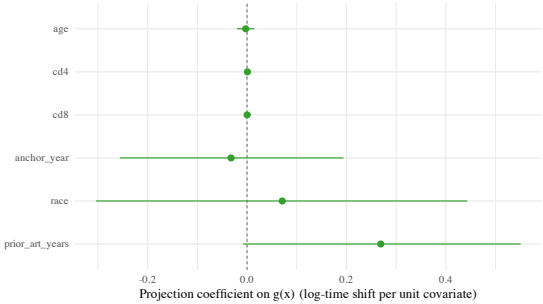
Method	Source	Learner	RMSE	Bias	Coverage	Width
Bayesian fusion forest	Both	—	**0.676**	**−0.026**	0.964	2.82
DNN	Trial	S	1.115	−0.185	0.375	0.95
		T	1.250	−0.247	0.913	3.73
	Pool	S	1.107	0.105	0.444	0.99
		T	1.350	0.137	0.991	4.92
XGBoost	Trial	S	1.161	−0.038	0.158	0.40
		T	1.593	−0.075	0.937	4.95
	Pool	S	1.114	0.151	0.269	0.64
		T	1.120	0.306	0.905	3.27
BJ-ELM	Trial	S	1.281	−0.068	0.445	1.42
		T	1.662	0.393	0.952	5.97
	Pool	S	1.282	0.309	0.300	0.90
		T	1.175	0.472	0.896	3.34
BJ-trees	Trial	S	1.209	−0.040	0.000	0.00
		T	1.354	−0.054	0.460	1.73
	Pool	S	1.222	0.028	0.000	0.00
		T	1.013	0.343	0.475	1.25

$\lambda_d = \lambda_u = 1$; both sources right-censored at $\approx 35\%$; 1000 replications.

How do the two baselines differ?

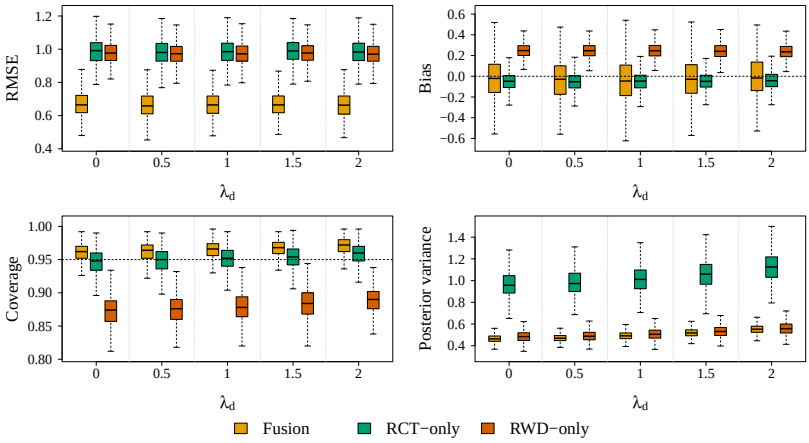
Linear projection of the deviation forest $d(x)$:

- Intercept 1.70 [1.31; 2.10].
- Prior ART years 0.27 [−0.01; 0.55].
- Age, CD4, CD8, year, race: all near zero.



The gap is essentially a constant shift — the cohort baseline sits well above the trial.

Between-source heterogeneity λ_d



All three estimators are flat in λ_d: the deviation forest absorbs the gap.